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Q-1: Explain the impact of selecting the K value over the different natured datasets.

Explanation:

The selection of an appropriate K (the number of nearest neighbors) has a significant effect on the performance of K-Nearest Neighbors (KNN) method, with the size and separation of classes in the dataset being among the factors determining the extent of that impact.

(A) Datasets With High Noise

- **Small K (e.g., K = 1 or 2):**
  - Very sensitive to noise.
  - A single noisy point can drastically affect prediction.
- **Large K:**
  - Helps in smoothing out noise.
  - Decisions are based on the majority among many points, reducing sensitivity to random noise.
  - Too large K, however, can **oversmooth**, causing misclassification.

Effect: Larger K is usually better for noisy datasets.

(B) Datasets With Well-Separated Classes

- **Small K:**
  - Works well because boundaries between classes are distinct.
  - Captures the detail of class separation accurately.
- **Large K:**
  - Still performs well but may include neighbors from other classes if K becomes too large.

Effect: Small moderate K produces good accuracy.

(C) Datasets With Overlapping Classes

- **Small K:**
  - May cause overfitting because the algorithm may pick neighbors from the overlapping region.
- **Large K:**
  - Reduces decision boundary complexity.
  - More stable but may misclassify minority classes in overlapping regions.

Effect: Moderate K values are preferred to avoid both overfitting and oversmoothing.

(D) High-Dimensional Datasets

- KNN suffers from the **curse of dimensionality**:
  - Distances grow less meaningful as dimensions increase.
- **Small K:**
  - Extremely unstable due to high dimensional noise.
- **Large K:**
  - Reduces instability but still impacted by the curse.

Effect: KNN may not perform well regardless of K, unless dimensionality reduction is applied.

(E) Imbalanced Datasets

- **Small K:**
  - Minority class predictions are difficult.
- **Large K:**
  - Majority class dominates the votes, causing bias.

Effect: Need techniques like weighted KNN or SMOTE.

Q-2: Explain the process of applying KNN in regression problems?

Explanation:

KNN is primarily known for classification but can also be used for **regression**.

Step-by-Step Process

Step 1: Prepare the Dataset

- Data must be numeric (or converted to numeric).
- Missing values must be handled.
- Scaling is essential (Min-Max, Standardization) because KNN uses distance metrics.

Step 2: Choose the Distance Metric

Common choices:

- Euclidean distance (most common)
- Manhattan distance
- Minkowski distance

Step 3: Choose K (Number of Neighbors)

- Small K → highly sensitive (high variance)
- Large K → smoother predictions (high bias)

Cross-validation is typically used to select the best K.

Step 4: Find K Nearest Neighbors for Each Query Point

- Compute the distance of the new point to all points in the training set.
- Select the **K closest points**.

Step 5: Predict the Output

In regression, the prediction is based on **aggregation of neighbors' values**, typically:

1. **Mean of K neighbors**

$$\hat{y} = \frac{1}{K}\sum_{i=1}^{K} \quad y_i$$

2. **Weighted average** (closer points get higher weight)

$$\hat{y} = \frac{\sum(w_i \cdot y_i)}{\sum w_i}$$

where

$$w_i = \frac{1}{distance_i}$$

Weighted KNN often improves performance.

Step 6: Evaluate the Model

Use metrics such as:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

Q-3: Can KNN be applied on the unlabelled (unsupervised learning) dataset? If yes, then how. If no, then why?

Explanation:

No, KNN can be applied on the unlabelled dataset since in KNN, we try to assign an unlabelled data point to an existing label in the dataset. Without a labelled dataset, we cannot give a class to that data point. Though, KNN can be applied in unsupervised learning since its nearest neighbor search can be used to generate new features for other machine learning models, to detect anomalies and outliers, and to preprocess the data in some cases, but not in classification itself.